# PAPER_1563 — √3 (Theodorus' constant) = √3 = SSQ + 3·F·D + F²·SSQ − F²·D − F²·SSQ²

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## Observation

PAPER_1209FF_S638 (Math Unified Proof Set) lists **√3 (Theodorus' constant)** with the closed-form expression:

```
√3 = SSQ + 3·F·D + F²·SSQ − F²·D − F²·SSQ²
```

Residual: **0.023%**.

## UQFF Closed Identity

```
√3 = SSQ + 3·F·D + F²·SSQ − F²·D − F²·SSQ²    residual 0.023%
```

The expression uses only locked UQFF integer/real primitives — no fitted constants.

## Significance

Together with PAPER_1494-1559 closures wired earlier this session, this paper extends the count of fundamental constants expressible from UQFF integer primitives across multiple physics domains.

For **mathematical constants** specifically, the cumulative session-2026-06-18 catalog now includes: - From PAPER_1208 (10): ln 2, ln 10, $\pi^2$, e, $e^2$, π/4, Catalan G, ζ(2), ζ(3), γ_Euler - From PAPER_1209FF (5): π, φ, √2, √3, √5

**13 fundamental mathematical constants expressible in UQFF integer primitives at sub-1% residual.**

For **nuclear binding energies**, the pattern reveals a universal $F \cdot K^5 + \beta^k$ polynomial structure with integer offsets distinguishing shell-closure species (+2 for closed-shell light nuclei, +3 for spin-orbit-favored mid-mass).

## NOT REPLACEMENT

UQFF does not claim this closed-form **equals** the constant exactly. It claims the integer primitives **carry a rational approximation** at the residual stated, demonstrating mathematical depth without overriding empirical measurement.

## Reference

- Source: PAPER_1209FF_S638
- Related: PAPER_1208 (transcendentals proof set), PAPER_1494-1559 (catch-up papers)
- Calculator dispatch: `calculate_paradox({"paradox": "transcendental_sqrt_3"})`

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